

Digital Image Processing:

Image Enhancement---Spatial domain[2]

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IMAGE ENHANCEMENT

1. Spatial domain methods

(1) Point-based methods - intensity transformations

- *Negative*
- *Log transformation*
- *Power-law*
- *Contrast stretching*
- *Gray-level slicing*
- *Histogram equalization*
- *Averaging*

(2) Neighborhood-based (Mask-based) methods spatial filter

2. Frequency domain methods

1 Fundamentals of spatial filtering

Filter:

- From frequency domain processing
- Accepting or rejecting
- For example: low-pass filter, high-pass filter

Here, we use words to represent the same thing:

- Spatial filter, masks, kernels, templates, windows
- A neighborhood-based spatial filter consists of:
 - ❑ Neighbourhood;
 - ❑ Predefined operation f

1 Fundamentals of spatial filtering

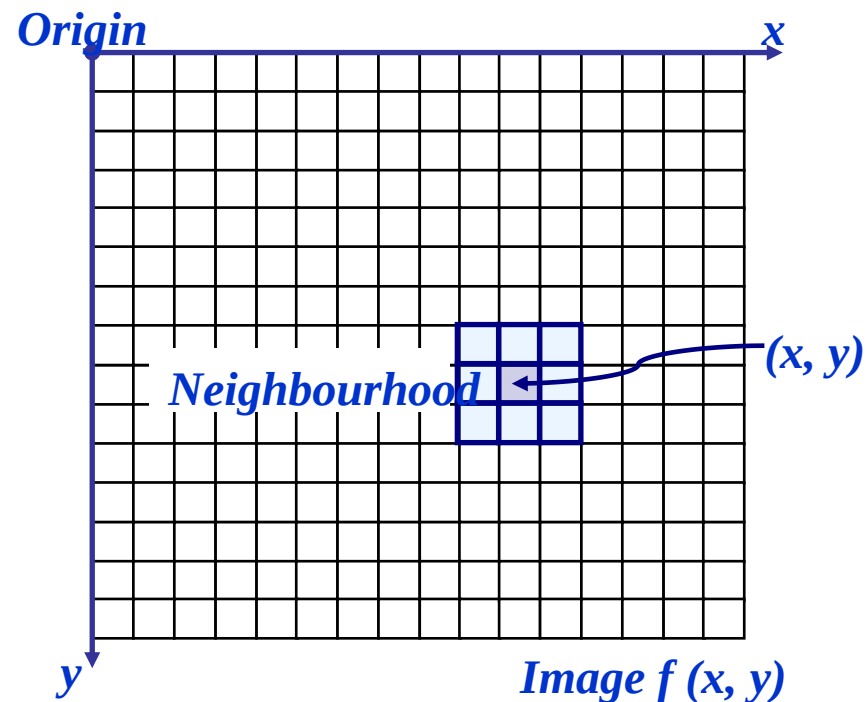
Neighborhood relations between pixels?

A pixel has 4 or 8 neighbors in 2D depending on the neighbor definition:

- 4-neighborhood: each neighbor must share an edge with the pixel.
- 8-neighborhood: each neighbor must share an edge or a **corner** with the pixel.
- Neighbourhoods are mostly a **rectangle around a central pixel**.
- **Any size** rectangle and **any shape** filter are possible.
3X3, 5X5, 7X7.....

1 Fundamentals of spatial filtering

- Neighbourhood operations operate on a **larger neighbourhood** of pixels than point operations



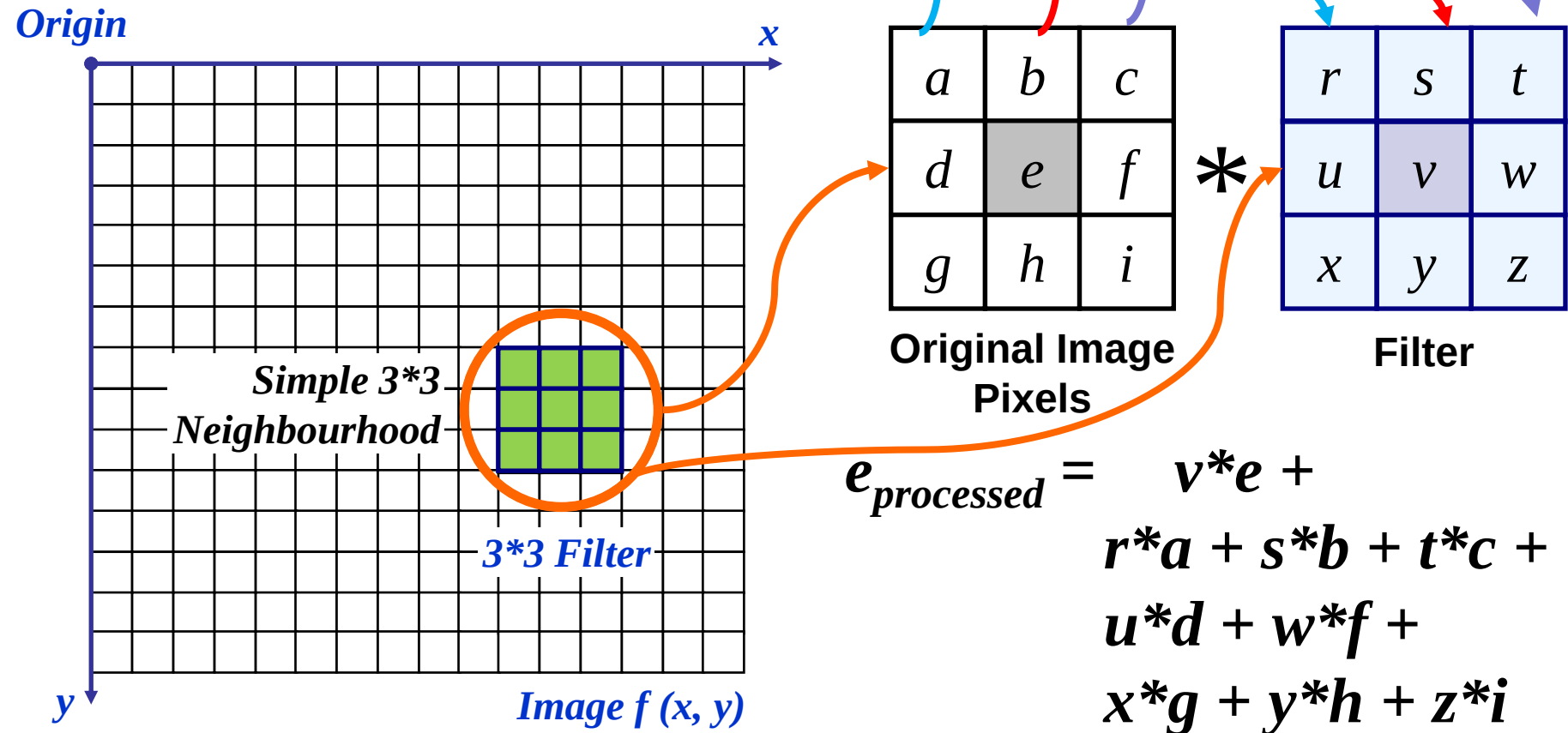
- Pixel value in output image is calculated from a local neighborhood in the input image
- Local neighborhood is described by a mask with a typical size of pixels

- ## Original Image

Enhanced Image

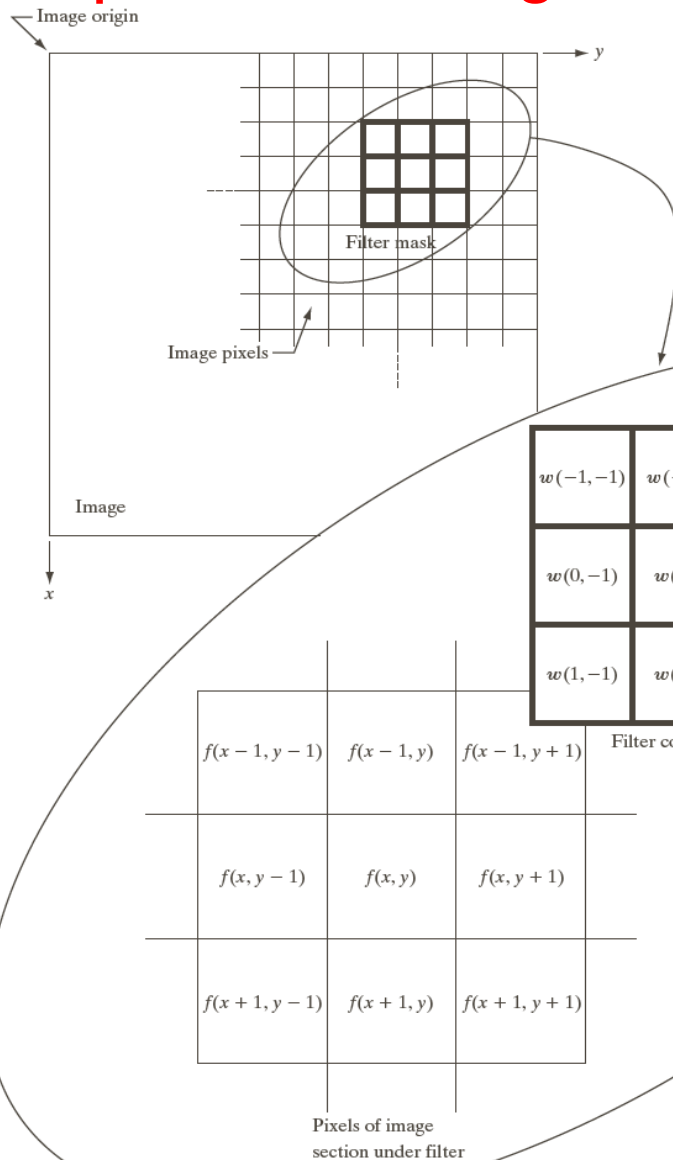
A 6x6 grid representing an enhanced image. The grid is labeled with 'x' and 'y' axes. The value '201' is written in red in the cell at row 3, column 3. Ellipses (...) are shown to the right and below the grid, indicating continuation.

1.1 The Spatial Filtering Process



The above is **repeated for every pixel** in the original image to generate the filtered image.

Spatial Filtering: Equation Form



Filtering can be given in equation form as:

$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x + s, y + t)$$

Notations are based on the image shown to the left.

Another representation of filter

W_1	W_2	W_3
W_4	W_5	W_6
W_7	W_8	W_9

1.2 Simple Neighborhood Operations

Two types:

- ① Linear spatial filter:
operation performed on the image pixels is linear;
 - ② Nonlinear spatial filter.
- **Min:** Set the pixel value to the minimum in the neighborhood
 - **Max:** Set the pixel value to the maximum in the neighborhood
 - **Median:** The median value of a set of numbers is the midpoint value in that set.

Original Image

123	127	128
140	145	148
133	154	183



Processed Image

Min:

Max:

Median:

	?	

2 Smoothing Spatial Filters

2.1 Nonlinear smoothing filter-Order Statistic Filters

Median filter

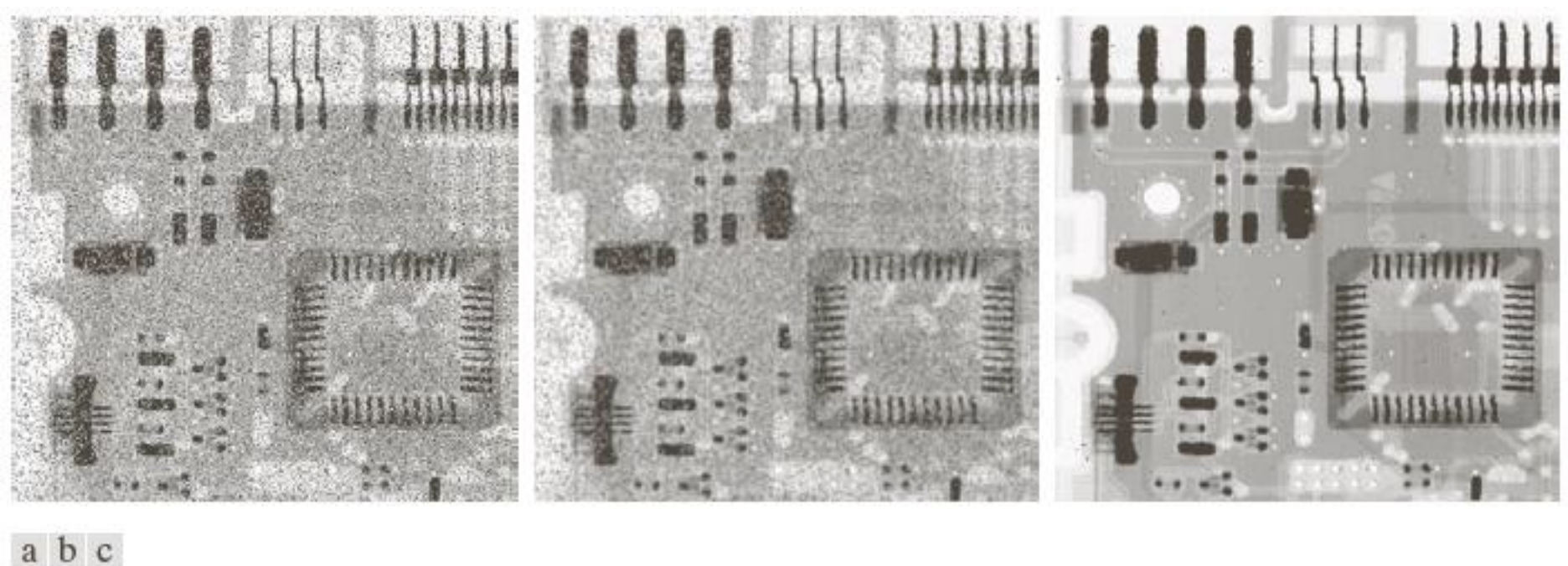


FIGURE 3.35 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3×3 averaging mask. (c) Noise reduction with a 3×3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Salt and pepper noise

2 Smoothing Spatial Filters

2.2 Smoothing Linear Filters

Smoothing operation:

- **Averaging:** One of the simplest spatial filtering operations:
 - ✓ **Average all of the pixels** in a neighbourhood around a central value.
 - ✓ It works for:
 - (a) **Noise reduction:** removed the noises.
 - (b) **Blurring:** removed small details from an image, bridging of small gaps in lines or curves.

2 Smoothing Spatial Filters

2.2 Smoothing Linear Filters

(1) Simple Averaging Filters

(2) Weighted Smoothing Filters

$$\frac{1}{9} \times$$

1	1	1
1	1	1
1	1	1

$$\frac{1}{16} \times$$

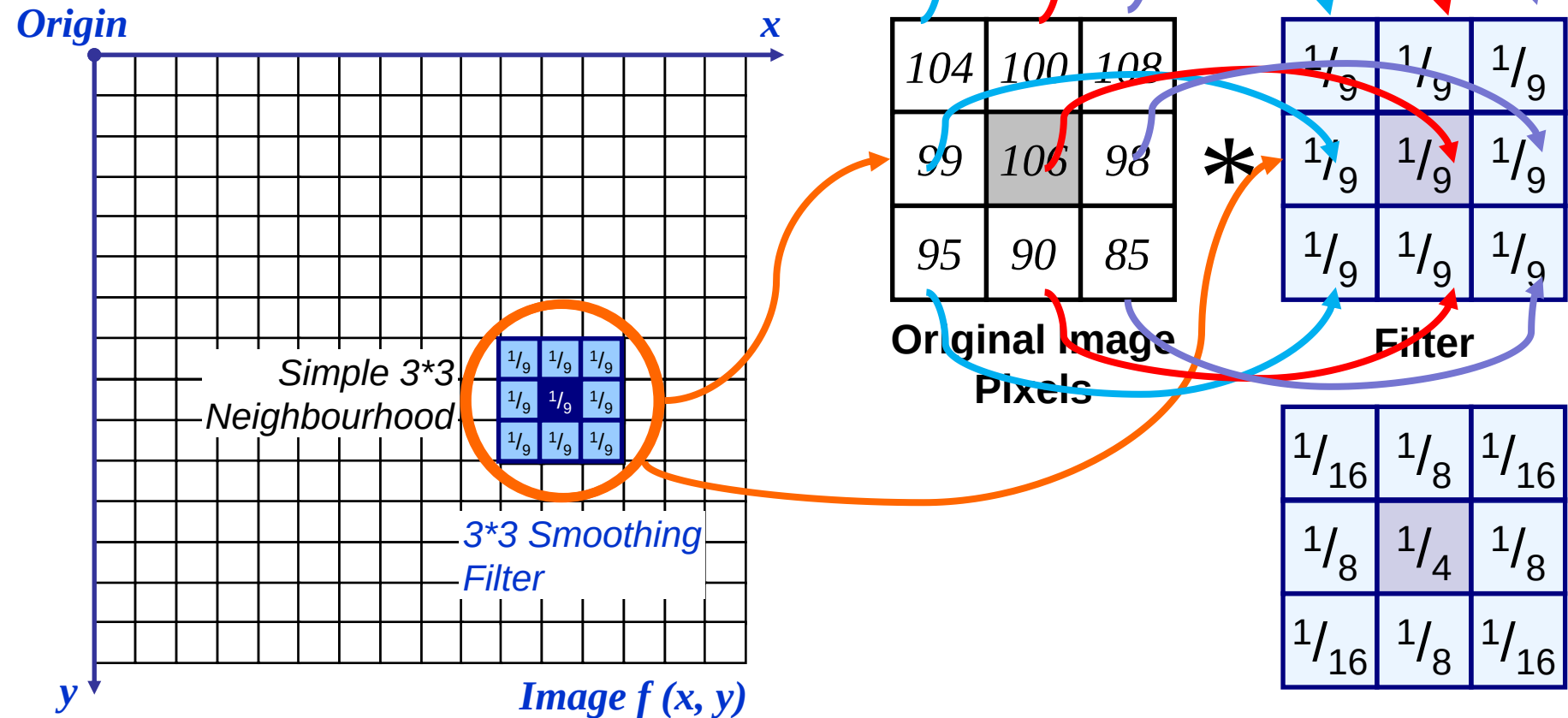
1	2	1
2	4	2
1	2	1

Pixels closer to the central pixel are more important!!!!

Digital Image Processing

2 Smoothing Spatial Filters

2.2 Smoothing Linear Filters



$$\begin{aligned}
 e &= \frac{1}{9} * 106 + \frac{1}{9} * 104 + \frac{1}{9} * 100 + \frac{1}{9} * 108 + \frac{1}{9} * 99 + \frac{1}{9} * 98 + \\
 &\quad \frac{1}{9} * 95 + \frac{1}{9} * 90 + \frac{1}{9} * 85 \\
 &= 98.3333
 \end{aligned}$$

2 Smoothing Spatial Filters

2.2 Smoothing Linear Filters

Original image

3*3

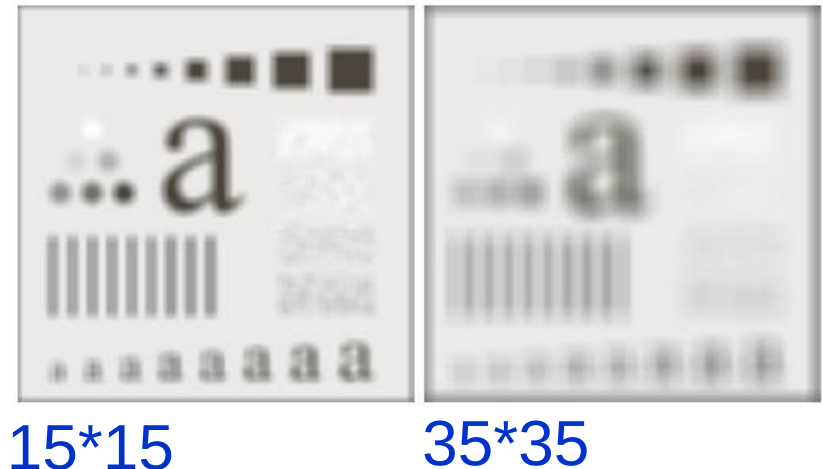
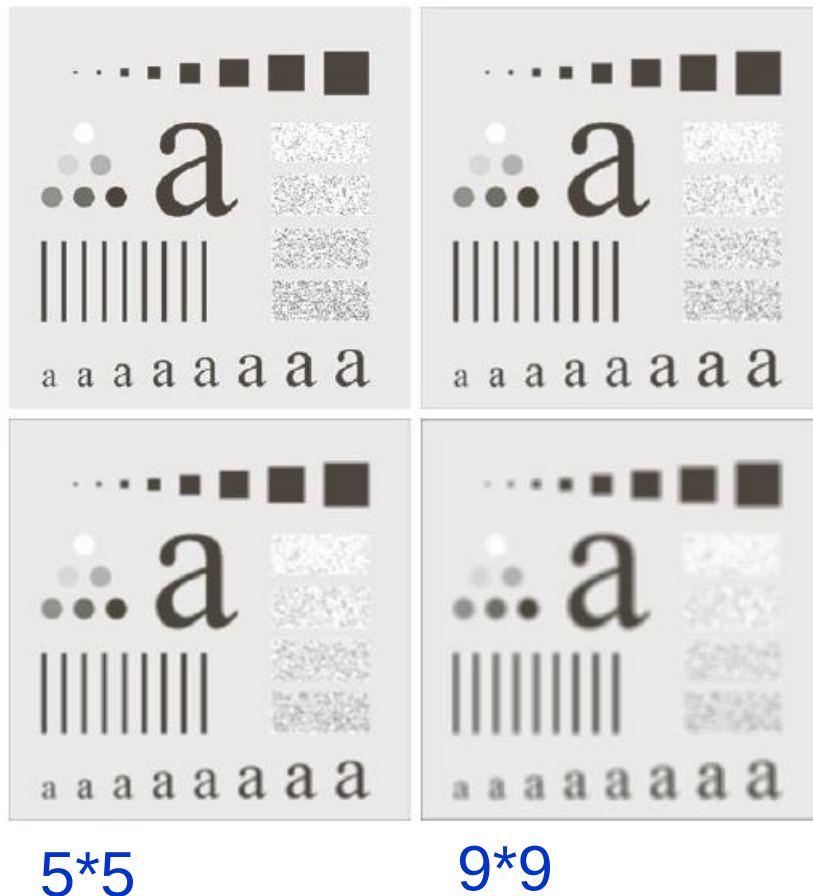
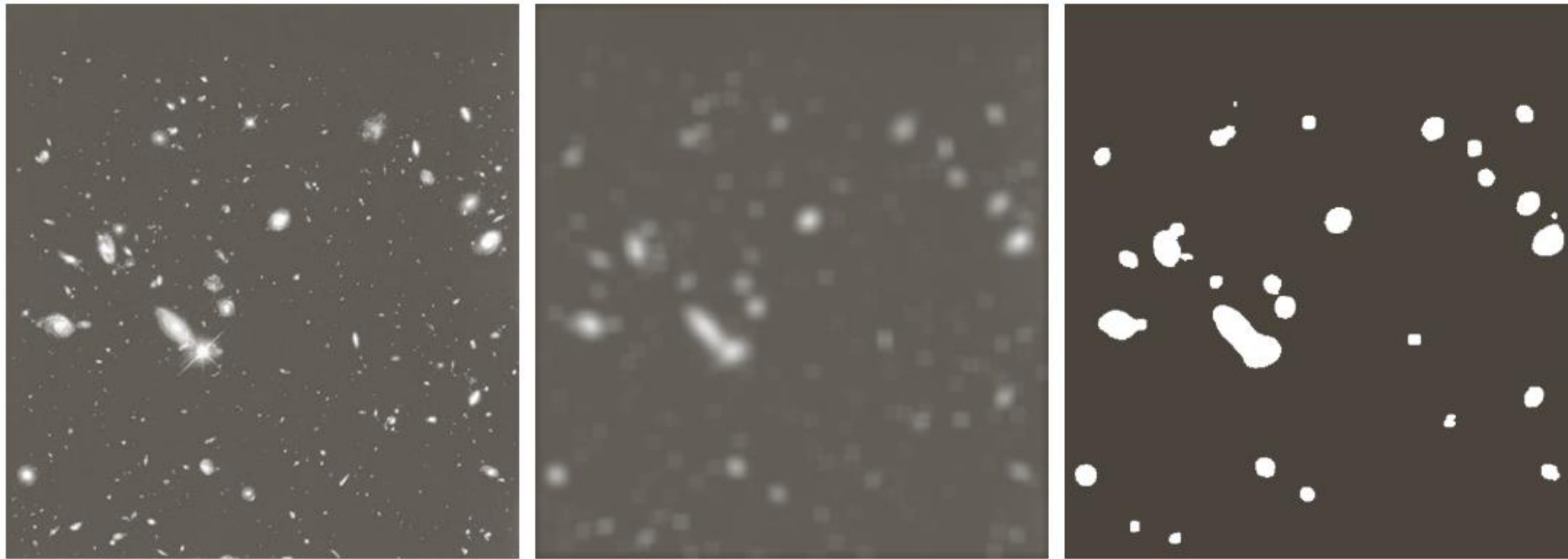
Blur the edge!!

FIGURE 3.33 (a) Original image, of size 500×500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $m = 3, 5, 9, 15$, and 35 , respectively. The black squares at the top are of sizes $3, 5, 9, 15, 25, 35, 45$, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their intensity levels range from 0% to 100% black in increments of 20% . The background of the image is 10% black. The noisy rectangles are of size 50×120 pixels.

a	b
c	d
e	f

2 Smoothing Spatial Filters

2.2 Smoothing Linear Filters

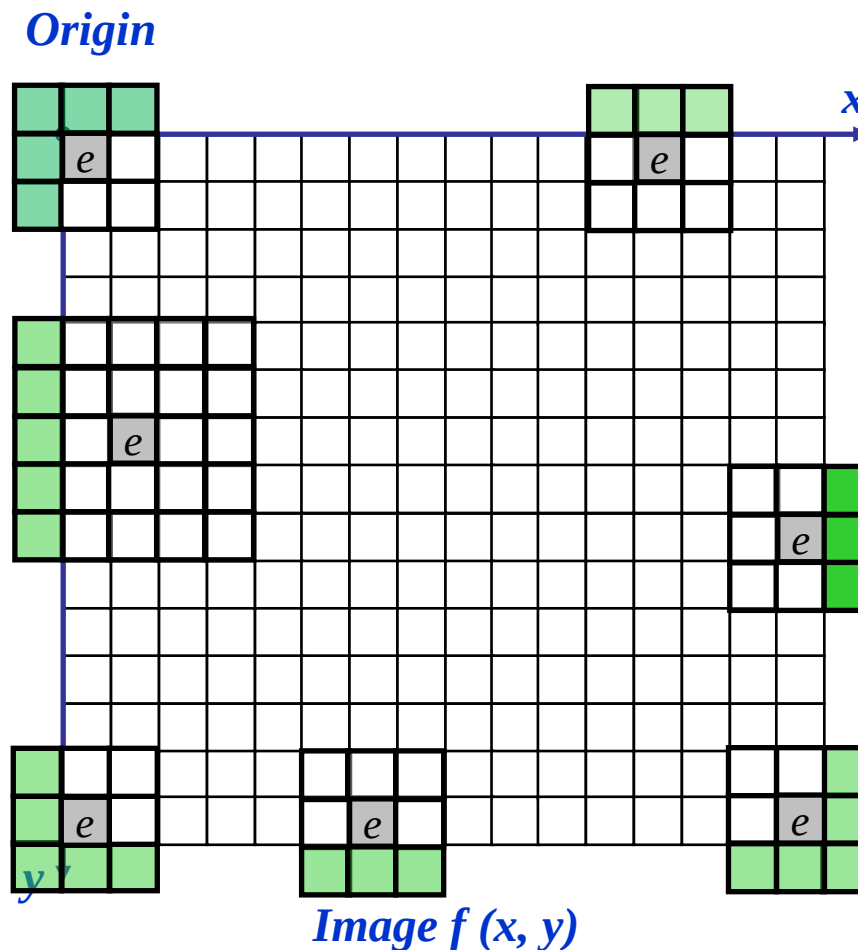


a b c

FIGURE 3.34 (a) Image of size 528×485 pixels from the Hubble Space Telescope. (b) Image filtered with a 15×15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)

Problem when filtering at the edges!

At the edges of an image we are missing pixels to form a neighbourhood.



How to solve?

Missing edge pixels:

1) Omit missing pixels

- Only works with some filters
- Add extra code and slow down processing

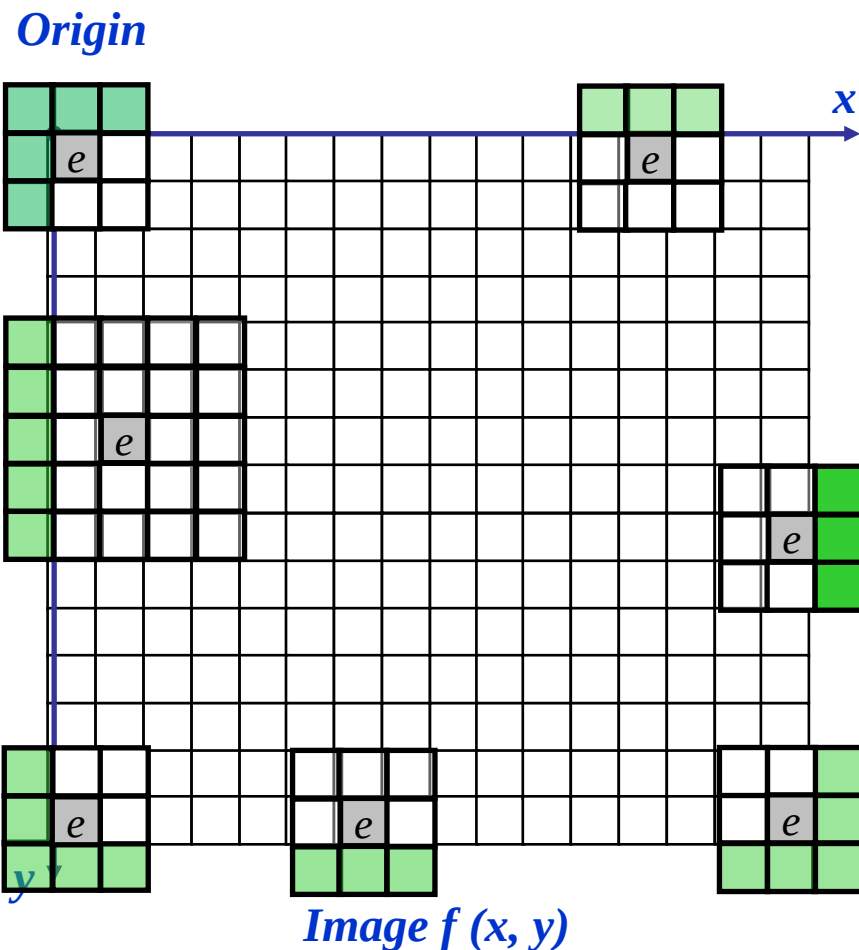
2) Pad the image

- Typically with either all white or all black pixels

3) Replicate border pixels

4) Truncate the image

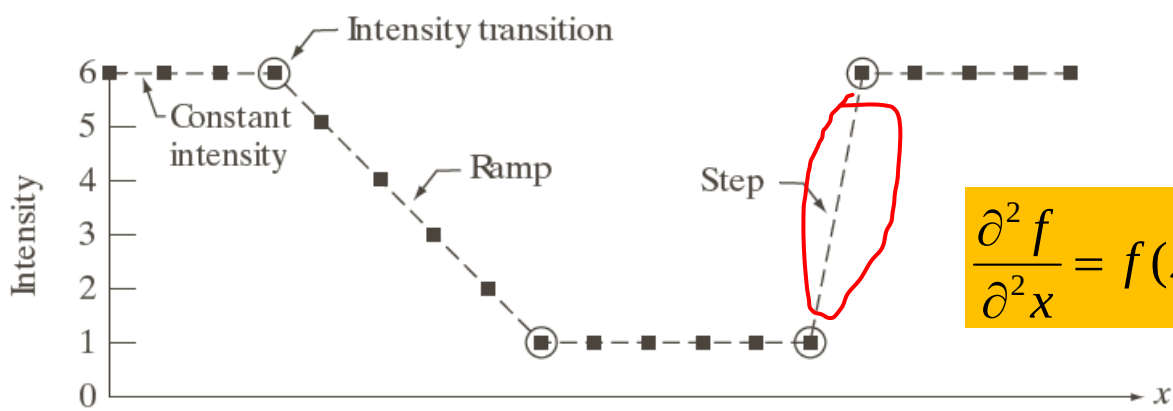
- Can cause some strange image artefacts



3 Sharpening Spatial Filters-Concepts

- Sharpening spatial filters highlight transitions in intensity, use for:
 - a) Highlight fine detail (edges)
 - b) Remove blurring from image
- Smoothing filters remove fine detail, analogous to integration. 积分
- Sharpening filters are based on *spatial differentiation*. 微分
- Differentiation measures the rate of change of a function

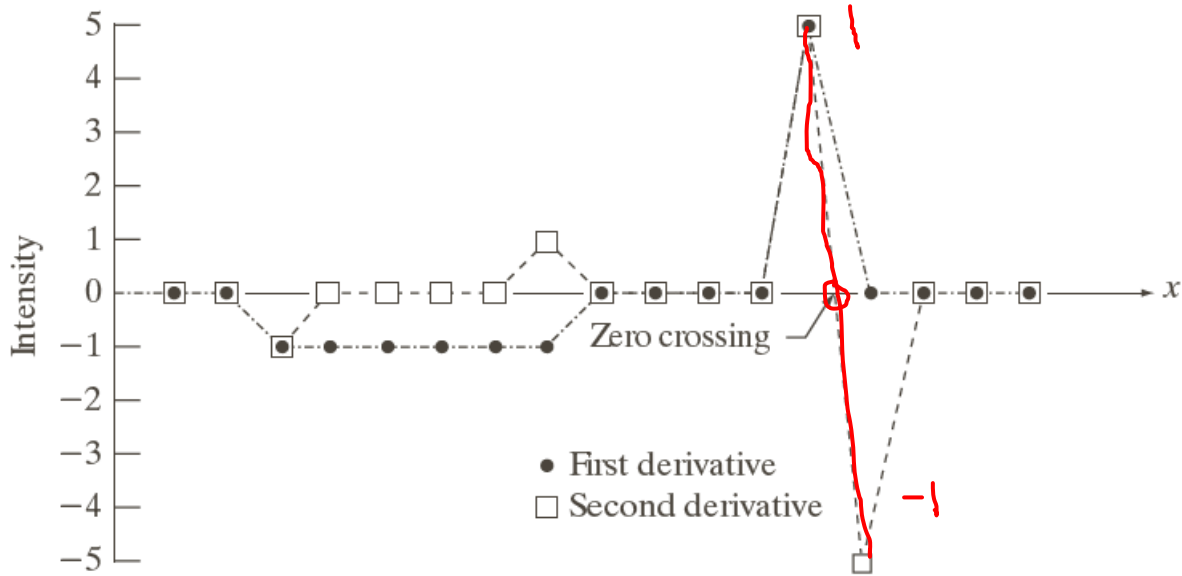
Spatial Differentiation



$$\frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

$$\frac{\partial^2 f}{\partial^2 x} = f(x + 1, y) + f(x - 1, y) - 2 f(x, y)$$

Scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
1st derivative	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	0	5	0	0	0	0	0
2nd derivative	0	0	-1	0	0	0	0	1	0	0	0	0	0	5	-5	0	0	0	0



a
b
c

FIGURE 3.36 Illustration of the first and second derivatives of a 1-D digital function representing a section of a horizontal intensity profile from an image. In (a) and (c) data points are joined by dashed lines as a visualization aid.

Spatial Differentiation—

1st Derivative

The formula for the 1st derivative of a function is as follows:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

It's the difference between subsequent values.

Used for: measure the rate of change of the function

Here, measure the rate of change of pixel intensity.

Spatial Differentiation—1st Derivative

For a function $f(x, y)$ the gradient of f at coordinates (x, y) is given as the column vector:

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

Change in x direction

Change in y direction

The **magnitude** of this vector is given by:

$$\begin{aligned} \nabla f &= \text{mag}(\nabla f) \\ &= [G_x^2 + G_y^2]^{1/2} = \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2} \end{aligned}$$

For practical reasons this can be **simplified as**:

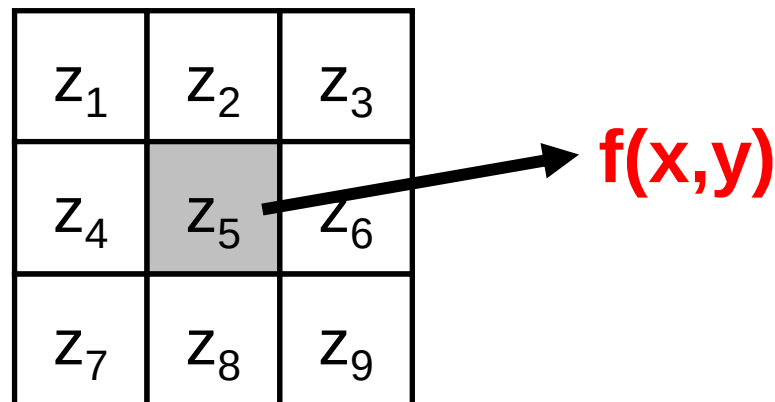
$$\nabla f \approx |G_x| + |G_y|$$

Spatial Differentiation—1st Derivative

There is some debate as to how best to **calculate these gradients** but we will use:

$$Z_6 - Z_5; \quad Z_8 - Z_5$$

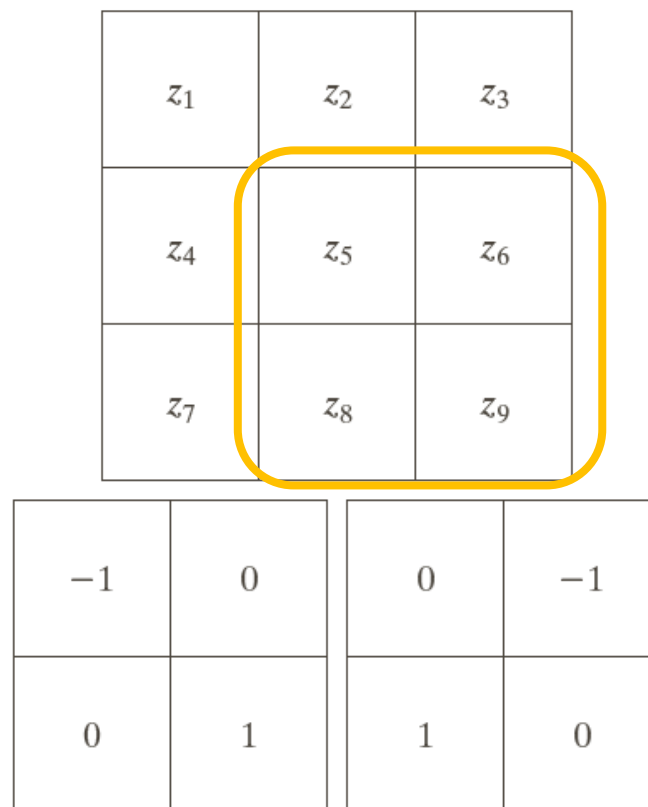
which is based on notions with following coordinates



Spatial Differentiation—1st Derivative

Operator 1: *Robert cross-gradient operators*

$$\nabla f \approx |z_9 - z_5| + |z_8 - z_6|$$



Spatial Differentiation—1st Derivative

Operator 2: *Sobel operator*

$$\nabla f \approx |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| \\ + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$$

-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

Spatial Differentiation—1st Derivative

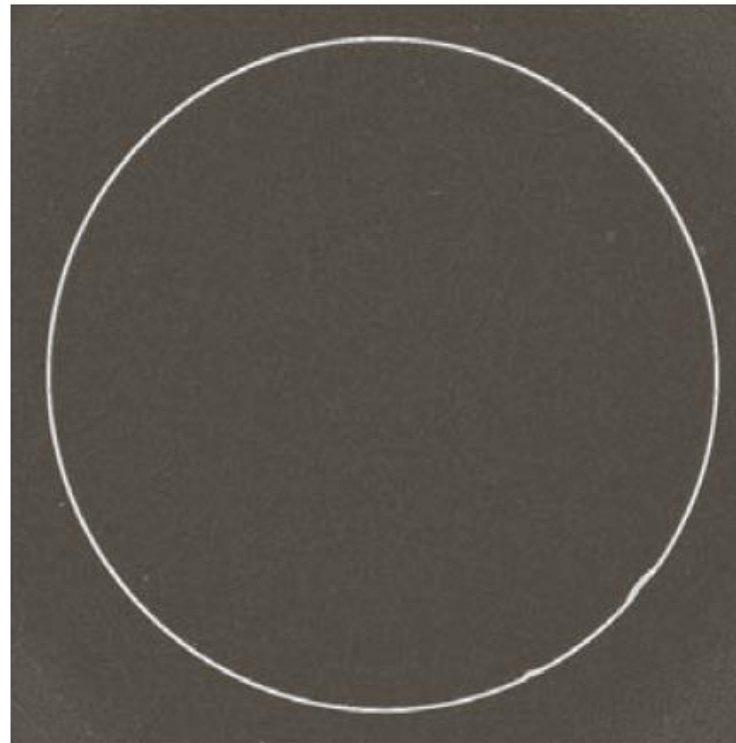
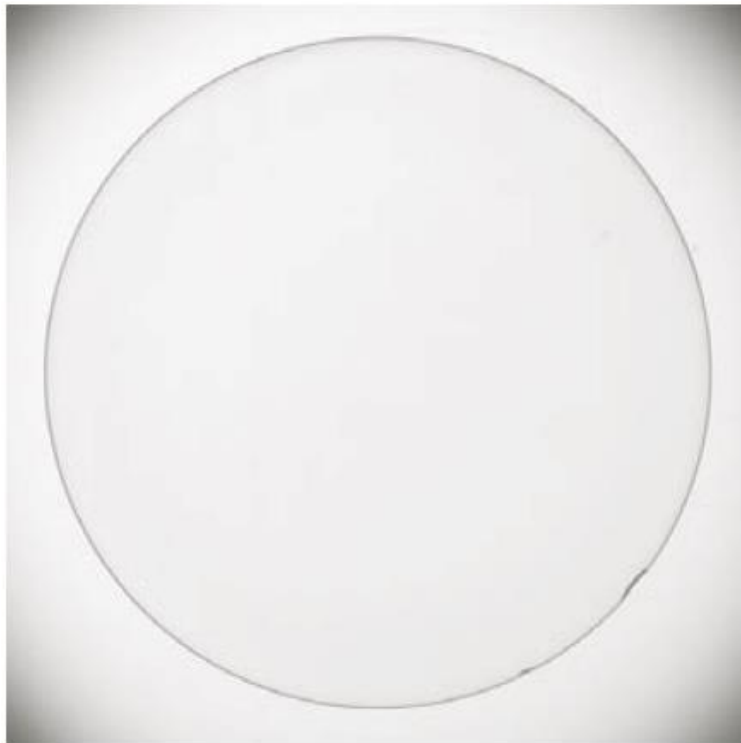
-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

Operator 2: *Sobel operator*

Question:

Why the sum of the all coefficients are 0?



a b

FIGURE 3.42
(a) Optical image of contact lens (note defects on the boundary at 4 and 5 o'clock).
(b) Sobel gradient.
(Original image courtesy of Pete Sites, Perceptics Corporation.)

Spatial Differentiation—2nd Derivative

The formula for the 2nd derivative of a function is as follows:

$$\frac{\partial^2 f}{\partial^2 x} = f(x+1) + f(x-1) - 2f(x)$$

Takes into account the values **both before and after** the current value

The 2nd derivative is more useful for image enhancement than the 1st

- Stronger response to fine detail
- Simpler implementation

For example: *Laplacian*

- One of the simplest sharpening filters
- Isotropic

Spatial Differentiation—2nd Derivative

-Laplacian

The **Laplacian** is defined as follows:

$$\nabla^2 f = \frac{\partial^2 f}{\partial^2 x} + \frac{\partial^2 f}{\partial^2 y}$$

where the partial 2nd order derivative in the x direction is defined as follows:

$$\frac{\partial^2 f}{\partial^2 x} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

and in the y direction as follows:

$$\frac{\partial^2 f}{\partial^2 y} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

Spatial Differentiation—2nd Derivative-Laplacian

The formula for Laplacian of two variables is as follows:

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)$$

a	b
c	d

FIGURE 3.37

(a) Filter mask used to implement Eq. (3.6-6).
 (b) Mask used to implement an extension of this equation that includes the diagonal terms.
 (c) and (d) Two other implementations of the Laplacian found frequently in practice.

0	1	0
1	-4	1
0	1	0

0	-1	0
-1	4	-1
0	-1	0

Isotropic for rotations of 90 degree

Spatial Differentiation—2nd Derivative-Laplacian

The formula for Laplacian of two variables is as follows:

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)$$

1	1	1	-1	-1	-1
1	-8	1	-1	8	-1
1	1	1	-1	-1	-1

Considering diagonal directions
Isotropic for rotations of 45 degree

Spatial Differentiation—2nd Derivative-Laplacian

Laplacian result highlights intensity discontinuities and deemphasizes regions with slowly intensity levels.

In practical, Sharpening an image===Subtract the Laplacian result from the original image to generate final sharpened enhanced image.

$$g(x, y) = f(x, y) - \nabla^2 f$$

Spatial Differentiation—2nd Derivative-Laplacian

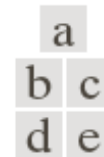
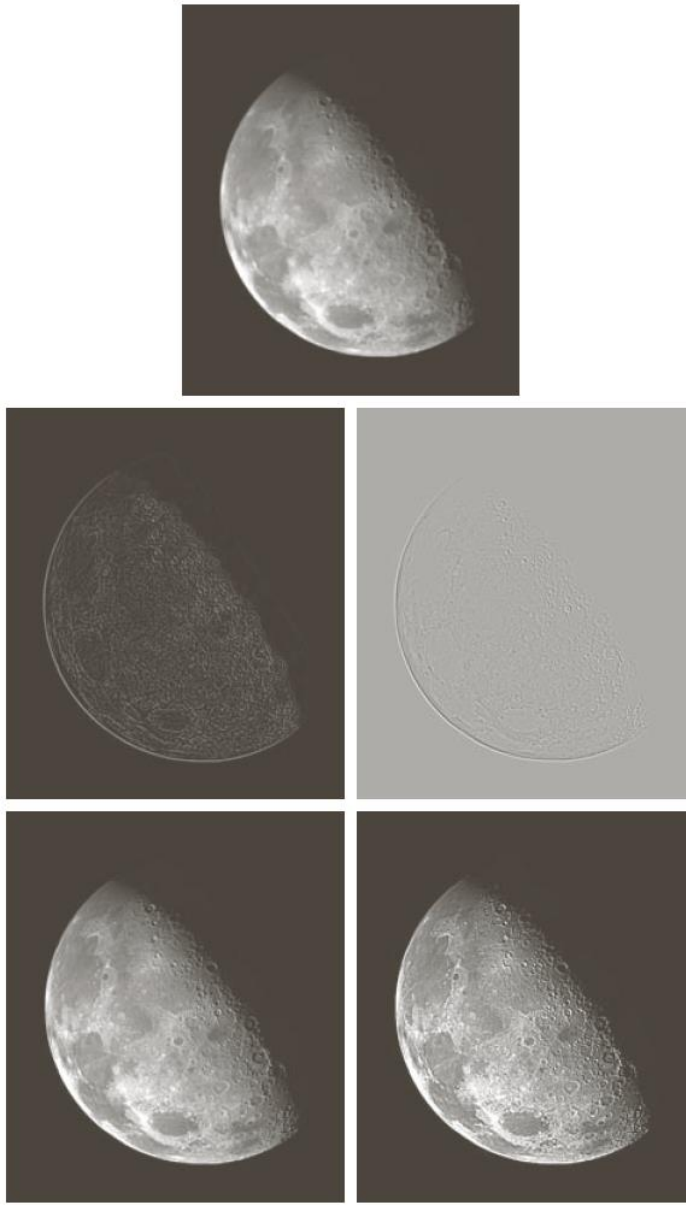


FIGURE 3.38

(a) Blurred image of the North Pole of the moon.

(b) Laplacian without scaling.

(c) Laplacian with scaling. (d) Image sharpened using the mask in Fig. 3.37(a).

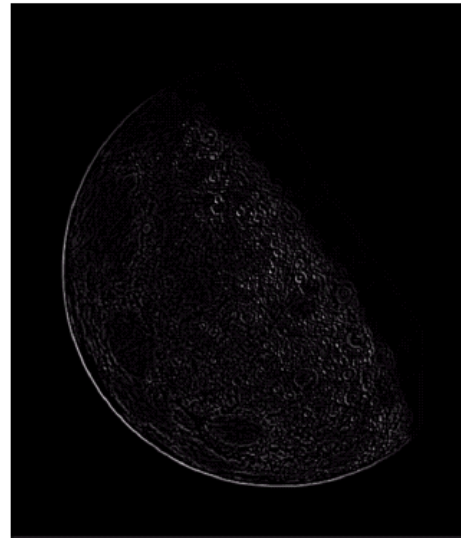
(e) Result of using the mask in Fig. 3.37(b).

(Original image courtesy of NASA.)

Spatial Differentiation—2nd Derivative-Laplacian



Original Image

Laplacian
Filtered Image

Sharpened Image

Enhancement! $g(x, y) = f(x, y) - \nabla^2 f$

In the final sharpened image edges and fine detail are much more obvious

Spatial Differentiation—2nd Derivative-Laplacian

One step enhancement:

Combines into a single filtering operation

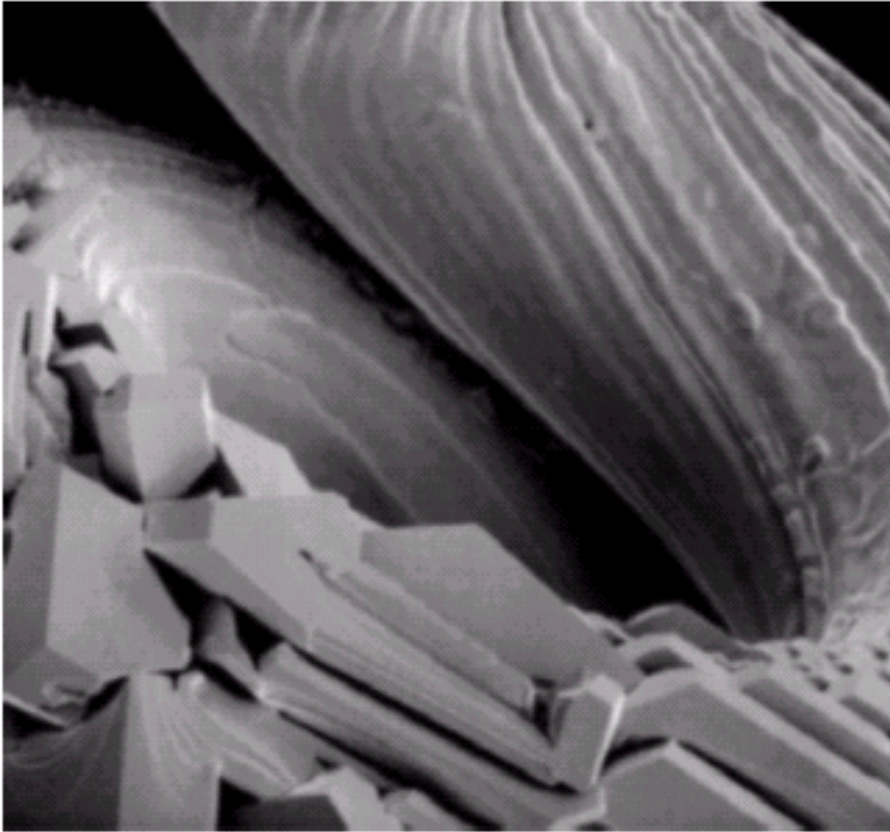
$$\begin{aligned} g(x, y) &= f(x, y) - \nabla^2 f \\ &= f(x, y) - [f(x+1, y) + f(x-1, y) \\ &\quad + f(x, y+1) + f(x, y-1) - 4f(x, y)] \end{aligned}$$

$$\begin{aligned} &= 5f(x, y) - f(x+1, y) - f(x-1, y) \\ &\quad - f(x, y+1) - f(x, y-1) \end{aligned}$$

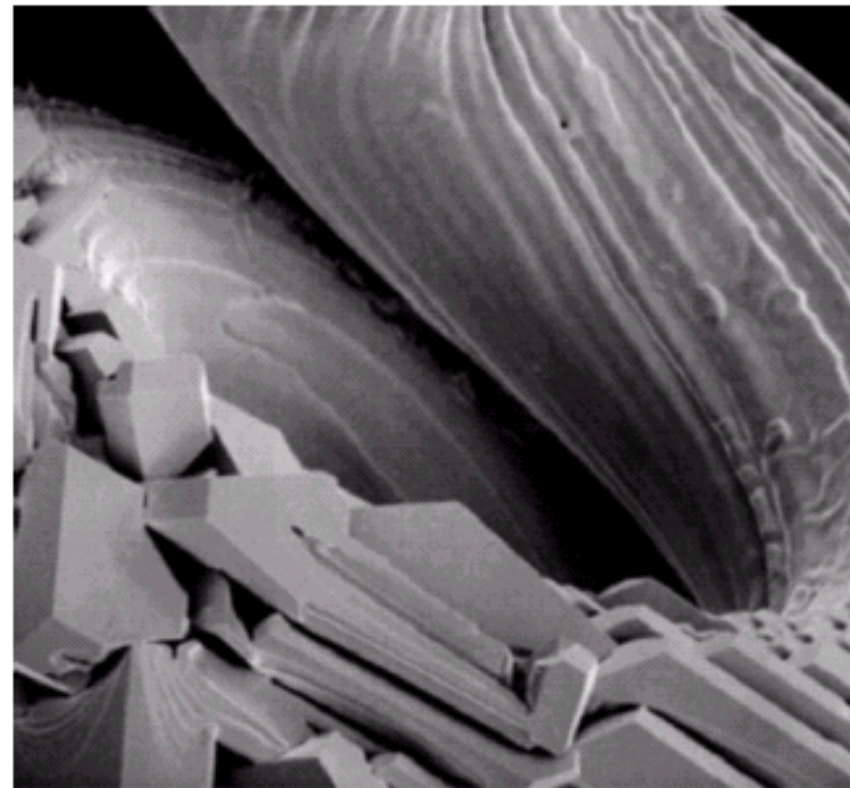
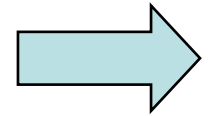
Simplified form!



0	-1	0
-1	5	-1
0	-1	0



0	-1	0
-1	5	-1
0	-1	0



Unsharp Masking and Highboost Filtering

Unsharp Masking: used in printing and publishing industry

- Blur the original image
- Create mask by subtracting the blurred image from the original image
- Add mask to the original

$$g_{mask}(x, y) = f(x, y) - f_s(x, y)$$

$$g(x, y) = f(x, y) + k * g_{mask}(x, y)$$

K=1: Unsharp Masking

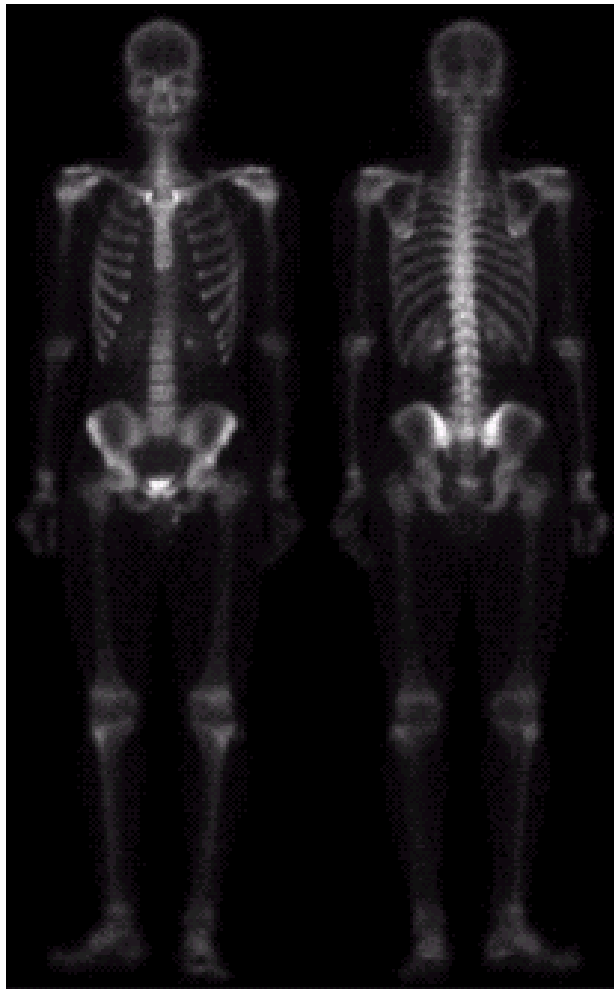
K>1: Highboost Filtering

1st & 2nd Derivatives

Comparing the 1st and 2nd :

- 1st order derivatives generally produce thicker edges
- 1st order derivatives have a stronger response to grey level step
- 2nd order derivatives produce a double response at step changes in grey level
- 2nd order derivatives have a stronger response to fine detail e.g. thin lines

Combining Spatial Enhancement Methods



(a)



(b)

Laplacian filter of bone scan (a)

Combining Spatial Enhancement Methods

(c)



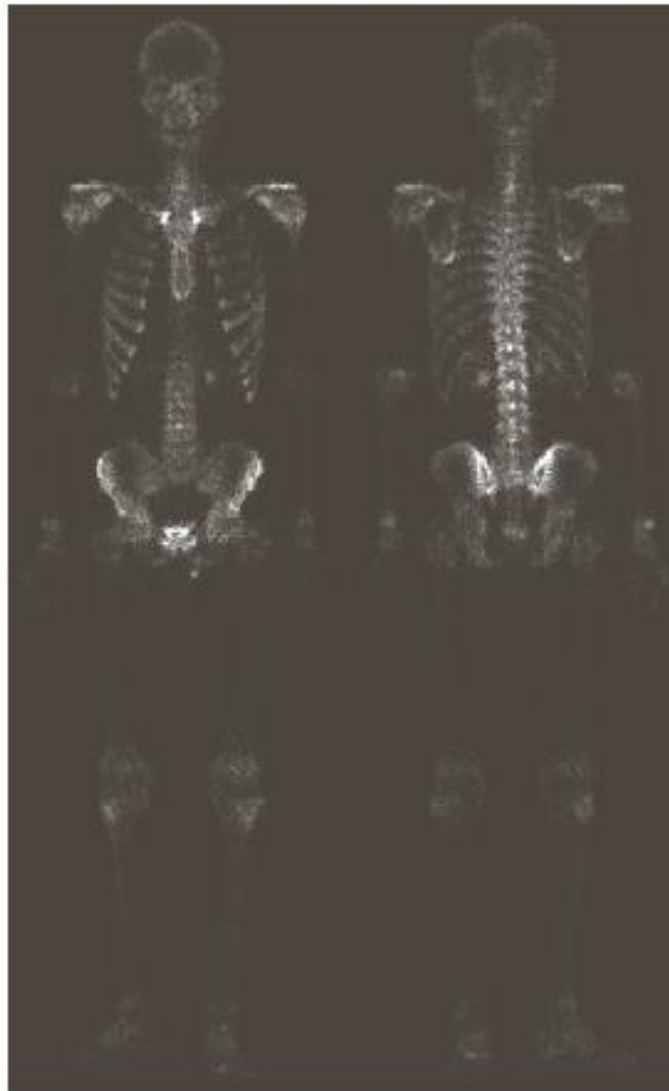
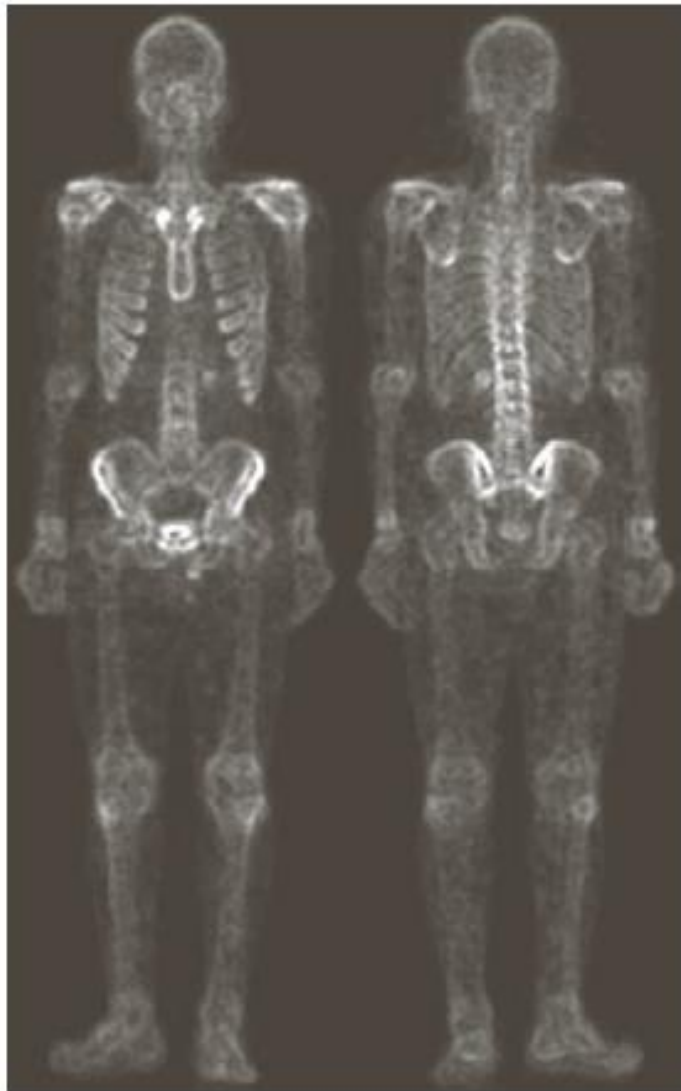
Sharpened version of bone scan achieved by subtracting (a) and (b)

(d)



Sobel filter of bone scan (a)

Combining Spatial Enhancement Methods



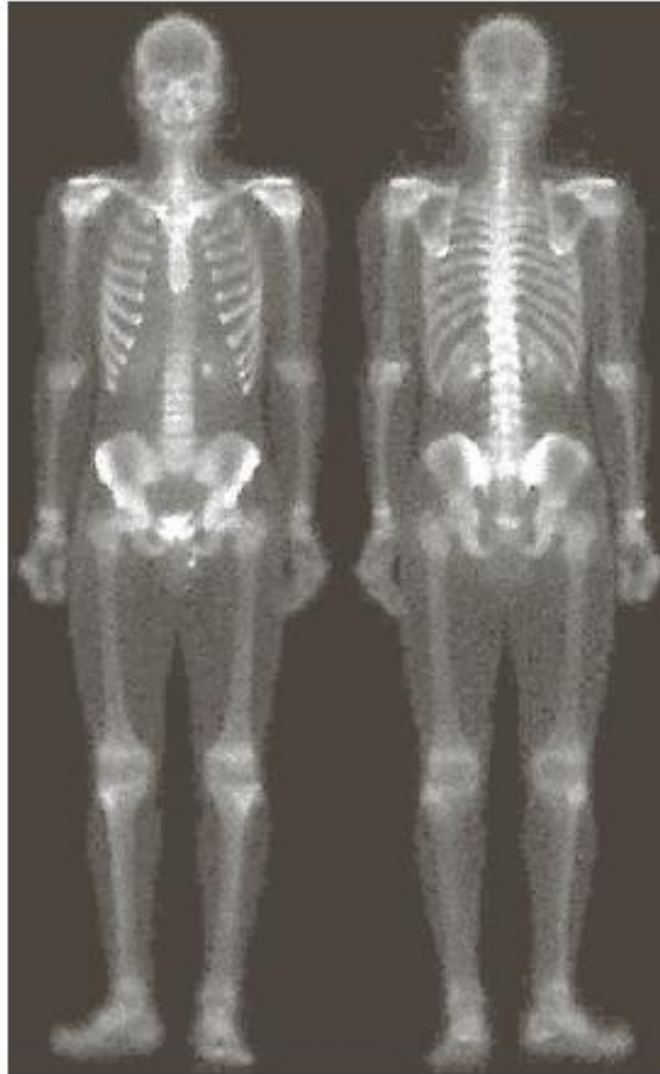
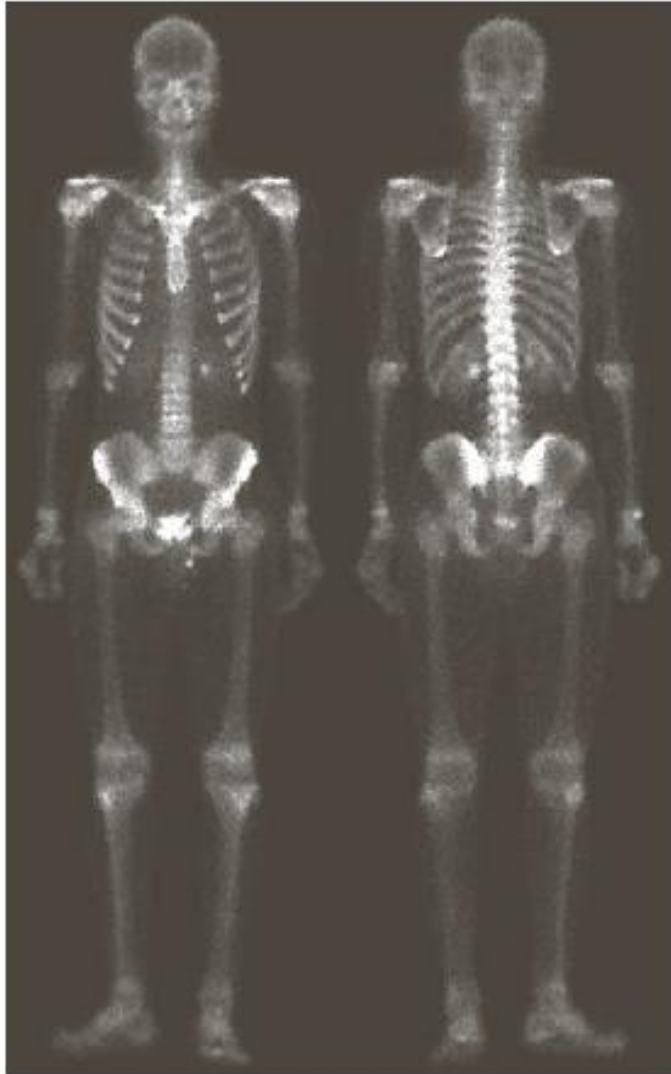
e	f
g	h

FIGURE 3.43

(Continued)

(e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e).

Combining Spatial Enhancement Methods



(g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)

original and final images



Questions:

- 1 What's the smoothing spatial filter?
- 2 What's the sharpening spatial filter?
- 3 Give an example of weighted average smoothing filter?
- 4 Give an example of Laplacian operator?
- 5 Give an example of Sobel operator?
- 6 Give an example of Robert operator?

Correlation & Convolution

- All filters above are *correlation* with the filter itself referred to as the *correlation kernel*
- *Convolution* is a similar operation

a	b	c
d	e	e
f	g	h

Original Image
Pixels

$*$

r	s	t
u	v	w
x	y	z

Filter

$$e_{processed} = v * e + \\ z * a + y * b + x * c + \\ w * d + u * e + \\ t * f + s * g + r * h$$

- For *symmetric filters* it makes no difference